

DESIGN AND SIMULATION OF A 1:8 ANALOG DEMULTIPLEXER

Using CMOS Transmission Gates in Cadence Virtuoso

Project Report

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Executive Summary

This project implements a 1:8 analog demultiplexer (DEMUX) as a three-stage binary tree of seven identical 1:2 analog DEMUX cells. Each 1:2 cell uses two CMOS transmission gates, and each transmission gate contains one NMOS and one PMOS in parallel. The full signal tree therefore contains 14 transmission gates, or 28 signal-path MOSFETs. Three external select bits S2, S1 and S0 choose one of eight outputs Y0-Y7.

The external digital-control requirement is 0 V / 2 V, while the analog signal must pass from -3.5 V to +3.5 V using ± 4 V supplies. Ideal VCVS blocks were therefore used inside the schematic as level-conversion/control drivers that generate complementary ± 4 V transmission-gate controls from the 0/2 V select inputs. The signal-switching core itself is transistor-level.

The final design was verified by functional transient simulation and by separate testbenches for on-resistance, input capacitance, turn-on time, bandwidth, charge injection and power. The measured worst-case on-resistance was approximately 24.51 Ω , input capacitance was tuned to approximately 55.07 pF, turn-on time was approximately 0.591 ns, worst measured charge injection was approximately 0.287 pC, and simulated average rail power was approximately 1.80 μ W. The measured -3 dB bandwidth remained above 1 GHz; the instructor explicitly allowed bandwidth greater than 1.2 MHz if an exact 1.2 MHz cutoff could not be obtained.

Important scope note: The power result is for the simulated DEMUX core with ideal VCVS level conversion. It does not include the power of a real transistor-level level shifter. The project is schematic/simulation only; layout, DRC, LVS and device-reliability sign-off are outside scope.

Report Structure

- 1. Project specifications and interpretation
- 2. Theory of the analog DEMUX and CMOS transmission gate
- 3. Circuit architecture and implementation
- 4. Functional operation and routing proof
- 5. Verification of every electrical criterion
- 6. Problems encountered and how they were solved
- 7. Final performance table, limitations and conclusion

1. Project Specifications

The given specification sheet is reproduced below. The wording uses “max” for several entries and gives ranges such as 50-65 pF and 1-20 pC. During the project, input capacitance was intentionally adjusted into the 50-65 pF band. For charge injection and power, the values are treated as maximum-limit requirements, so a smaller measured value is favorable.

Name of the Parameter	Value	Unit
Supply Voltage	+/-4	V
Logic High Level	2	V
Logic Low Level	0	V
Input Capacitance (max)	50 – 65	pF
Charge Injection over the full signal swing range (max)	1 – 20	pC
Switching On time (t_{ON}) (at $R_L = 1k\Omega$, $C_L = 10$ pF) (max)	120	ns
Analog Signal Range	-3.5 to 3.5	V
Power Dissipation (max)	1 – 25	mW
On-Resistance	10 – 25	Ω
Bandwidth, -3 dB	1.2	MHz

Figure 1. Original project specification table.

Parameter	Required value	Unit	Design interpretation
Supply voltage	± 4	V	Fixed rails
Logic high	2	V	External select input
Logic low	0	V	External select input
Input capacitance	50-65	pF	Final design targeted inside this band
Charge injection (max)	1-20	pC	Upper-limit interpretation; worst case compared with 20 pC
Turn-ON time (max)	120	ns	$R_L = 1$ k Ω , $C_L = 10$ pF
Analog signal range	-3.5 to +3.5	V	Full-swing operation
Power dissipation (max)	1-25	mW	Compared with 25 mW upper limit
ON resistance	10-25	Ω	Full 3-stage selected path
Bandwidth, -3 dB	1.2	MHz	Instructor preferred actual cutoff; fallback accepted: >1.2 MHz

2. Theory and Operating Principle

2.1 What an analog demultiplexer does

An analog demultiplexer routes one analog input to exactly one of several outputs according to a digital select word. Unlike a purely digital DEMUX, the selected path must pass a continuously varying voltage, including positive and negative values, with low distortion and low series resistance. The unselected outputs should be isolated as strongly as practical.

For a 1:8 DEMUX, three select bits are sufficient because $2^3 = 8$. The binary value $S_2S_1S_0$ identifies the selected output.

S2 S1 S0	Selected output
000	Y0
001	Y1
010	Y2
011	Y3
100	Y4
101	Y5
110	Y6
111	Y7

2.2 CMOS transmission gate as the analog switch

A CMOS transmission gate places an NMOS and a PMOS in parallel between the same two signal nodes. Complementary gate controls make both devices conduct at the same time. The NMOS passes low and mid-level voltages efficiently, while the PMOS improves conduction near the positive end of the signal range. Using both devices therefore gives a lower and more uniform switch resistance across the analog swing than either device alone.

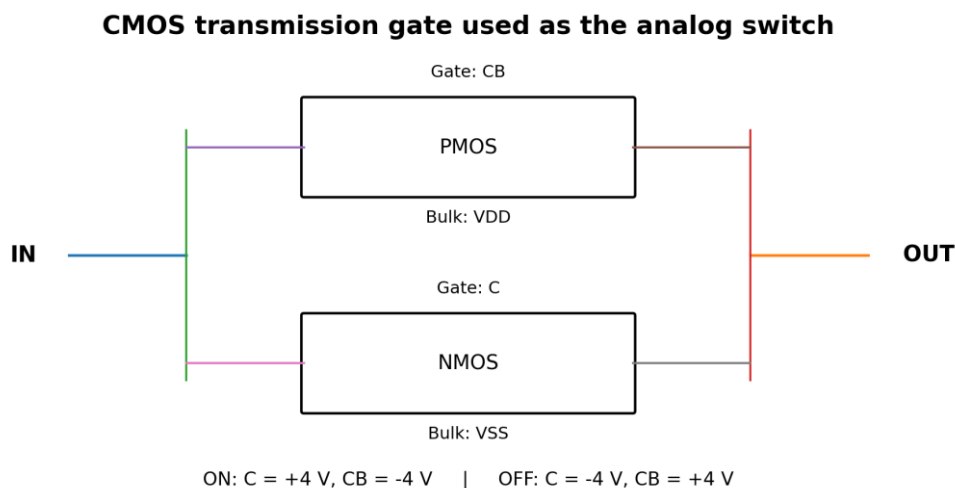


Figure 2. Conceptual CMOS transmission gate used as the bidirectional analog switch.

In the implemented `tg_cell`, the NMOS gate is C and the PMOS gate is CB. The MOS body terminals are tied to the supply rails: NMOS bulk to VSS and PMOS bulk to VDD.

ON state: C = +4 V, CB = -4 V

OFF state: C = -4 V, CB = +4 V

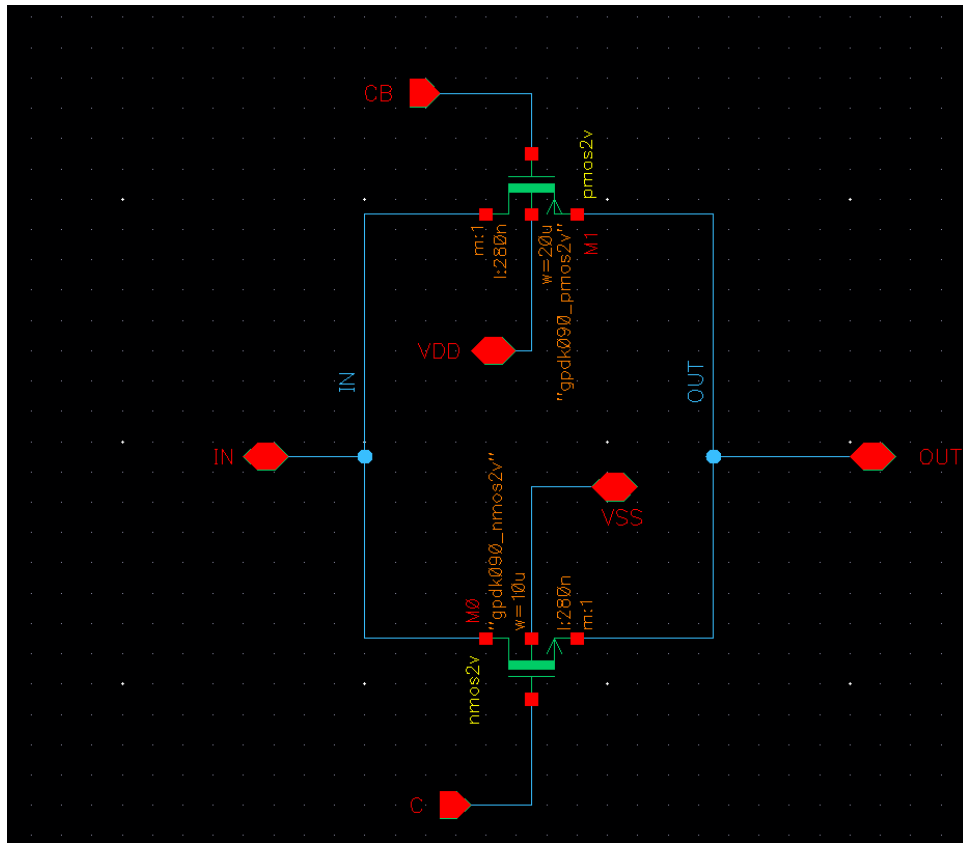


Figure 3. Transmission-gate cell topology captured during implementation; device widths were optimized later for the final resistance target.

Final switch sizing used for the verified design: NMOS total width $\approx 100\text{ }\mu\text{m}$ ($5 \times 20\text{ }\mu\text{m}$), PMOS total width $\approx 180\text{ }\mu\text{m}$ ($6 \times 30\text{ }\mu\text{m}$), and channel length $L = 280\text{ nm}$. The larger PMOS width compensates for lower hole mobility and helped keep the complete three-stage path below the $25\text{ }\Omega$ on-resistance limit.

2.3 Building one 1:2 analog DEMUX from two transmission gates

A 1:2 analog DEMUX is formed by connecting two transmission gates to the same input. Only one transmission gate is ON at a time. When S is low, OUT0 is selected; when S is high, OUT1 is selected. The complementary input SB guarantees that the two switches receive opposite commands.

S = LOW → TG0 ON, TG1 OFF → IN routed to OUT0

S = HIGH → TG0 OFF, TG1 ON → IN routed to OUT1

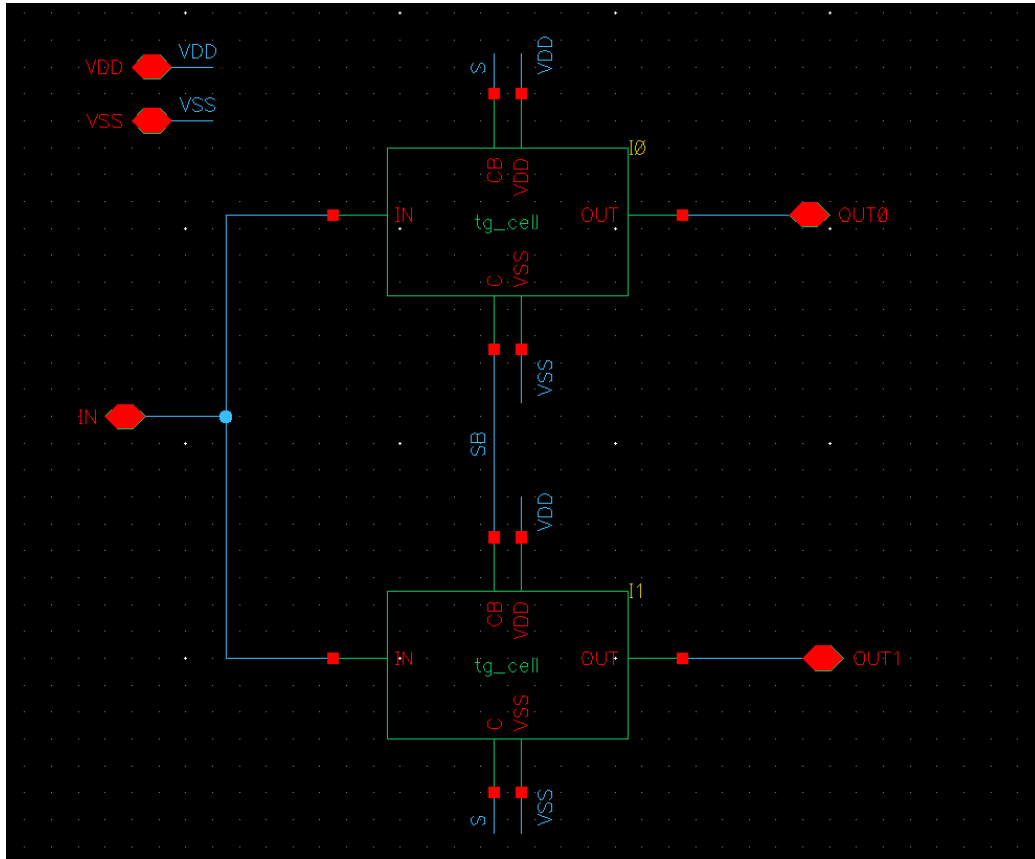


Figure 4. Implemented 1:2 analog DEMUX cell using two tg_cell blocks.

2.4 Why seven 1:2 DEMUX cells make one 1:8 DEMUX

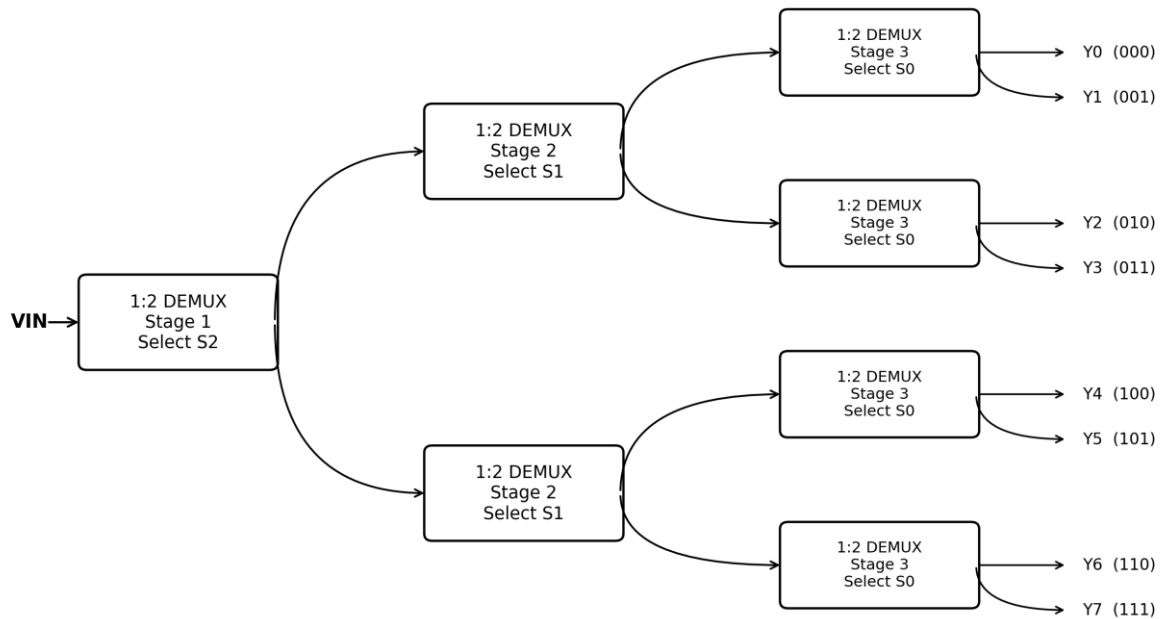
The 1:8 design is a complete binary tree with three decision stages. Stage 1 uses one 1:2 DEMUX and is controlled by the most-significant bit S2. Stage 2 uses two 1:2 DEMUX cells controlled by S1. Stage 3 uses four 1:2 DEMUX cells controlled by S0. The number of cells is therefore $1 + 2 + 4 = 7$.

$$\text{Number of 1:2 cells} = 1 + 2 + 4 = 7$$

$$\text{Number of signal transmission gates} = 7 \times 2 = 14$$

$$\text{Number of signal MOSFETs} = 14 \times 2 = 28$$

1:8 Analog DEMUX as a 3-stage binary tree



Number of 1:2 cells = $1 + 2 + 4 = 7$; one complete path contains three transmission-gate stages.

Figure 5. Binary-tree explanation of the seven-cell cascade and output selection.

Every selected analog signal travels through exactly three 1:2 stages, and therefore through three ON transmission gates in series. This is why the single-switch resistance had to be designed low enough that the total path would remain within the 10-25 Ω project range.

2.5 Converting the 0/2 V select inputs to ± 4 V switch controls

The external select specification is 0 V for logic LOW and 2 V for logic HIGH. The transmission-gate core was driven with complementary -4 V / $+4$ V controls. To bridge these voltage domains in the schematic-only project, ideal analogLib VCVS elements generate one direct and one complementary control for each select bit.

$$S_H = 4S - 4$$

$$S_HB = 4 - 4S$$

Therefore, $S = 0$ V produces $S_H = -4$ V and $S_HB = +4$ V, while $S = 2$ V produces $S_H = +4$ V and $S_HB = -4$ V. This gives the transmission gates the required complementary drive while keeping the external interface at 0/2 V.

Ideal level conversion used in the schematic-only implementation

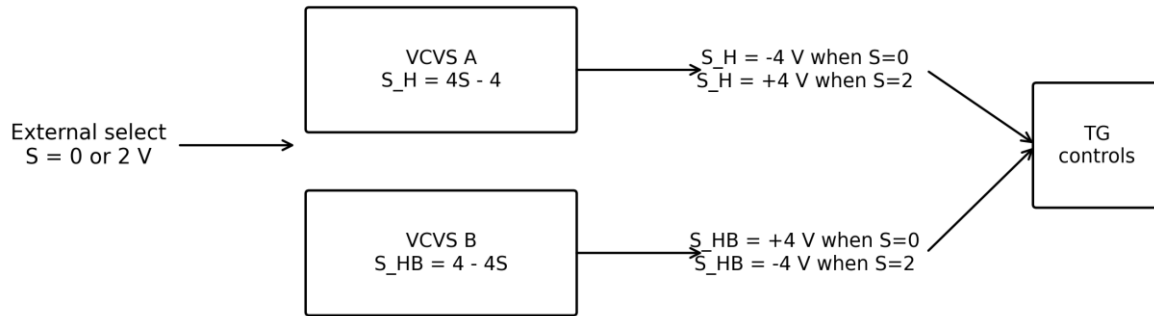


Figure 6. Ideal 0/2 V to ± 4 V complementary control conversion used internally.

Modeling limitation: The ideal VCVS level converters validate logic/control behavior but do not represent a physical level-shifter's power, delay, area or reliability. This is explicitly accounted for when interpreting the power result.

3. Circuit Architecture and Cadence Implementation

3.1 Hierarchical cells

- `tg_cell`: one NMOS + one PMOS in parallel, with IN, OUT, C, CB, VDD and VSS pins.
- `demux_1to2`: two `tg_cell` instances sharing IN; OUT0 is active for LOW select and OUT1 for HIGH select.
- `demux_1to8`: seven `demux_1to2` instances arranged as a 1-2-4 tree, plus internal VCVS control conversion and the final input-capacitance tuning capacitor.

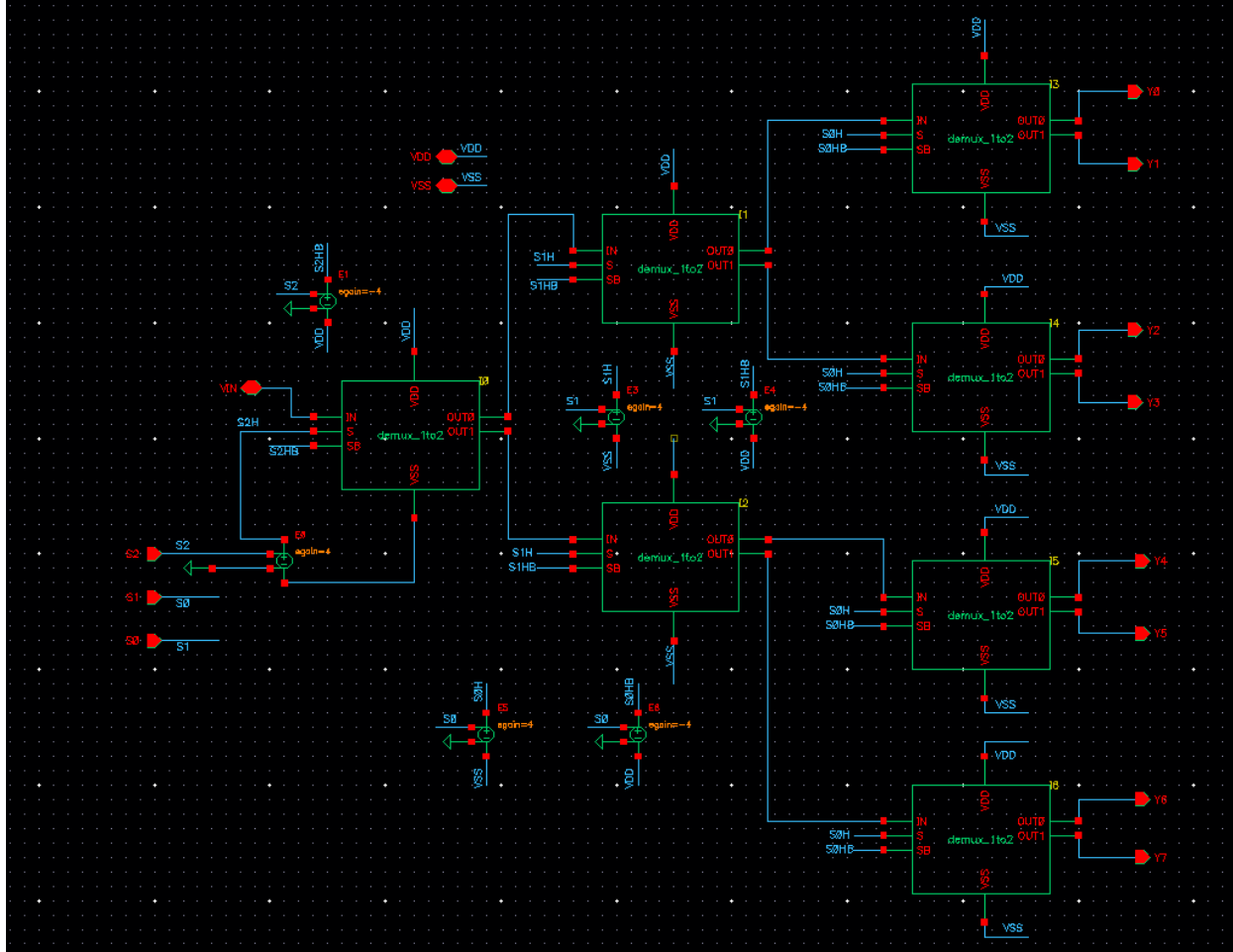


Figure 7. Internal Cadence architecture showing the seven 1:2 cells and control-level conversion. The final VIN capacitance-tuning capacitor was added after this architecture capture.

Analog-path connections follow the binary tree: I0 splits VIN into the upper and lower halves; I1/I2 split those halves; I3-I6 create Y0-Y7. Stage controls are S2, S1 and S0 respectively. Matching control assignment across an entire stage preserves the binary mapping shown earlier.

3.2 Input-capacitance tuning in the final block

The first AC-current measurement showed an intrinsic input capacitance of only about 2.05 pF. Because the project target was being treated as a 50-65 pF design band, approximately 53 pF was added from VIN to VSS inside the final `demux_1to8` block. Repeating the same measurement gave 55.07 pF. If the specification were interpreted strictly only as a maximum, the original 2.05 pF already satisfied it; the final design was intentionally tuned into the stated band.

$$C_{IN,final} \approx C_{intrinsic} + C_{added} \approx 2.05 \text{ pF} + 53 \text{ pF} \approx 55 \text{ pF}$$

3.3 Functional testbench

The main functional testbench uses a 100 kHz sine input with 3.5 V peak amplitude. The select inputs are three binary pulse sources with periods 20 μ s, 40 μ s and 80 μ s so that every binary code appears once during an 80 μ s simulation. For a clean routing demonstration, 10 k Ω pull-downs were placed at Y0-Y7. These resistors are testbench loads; they are not part of the DEMUX core.

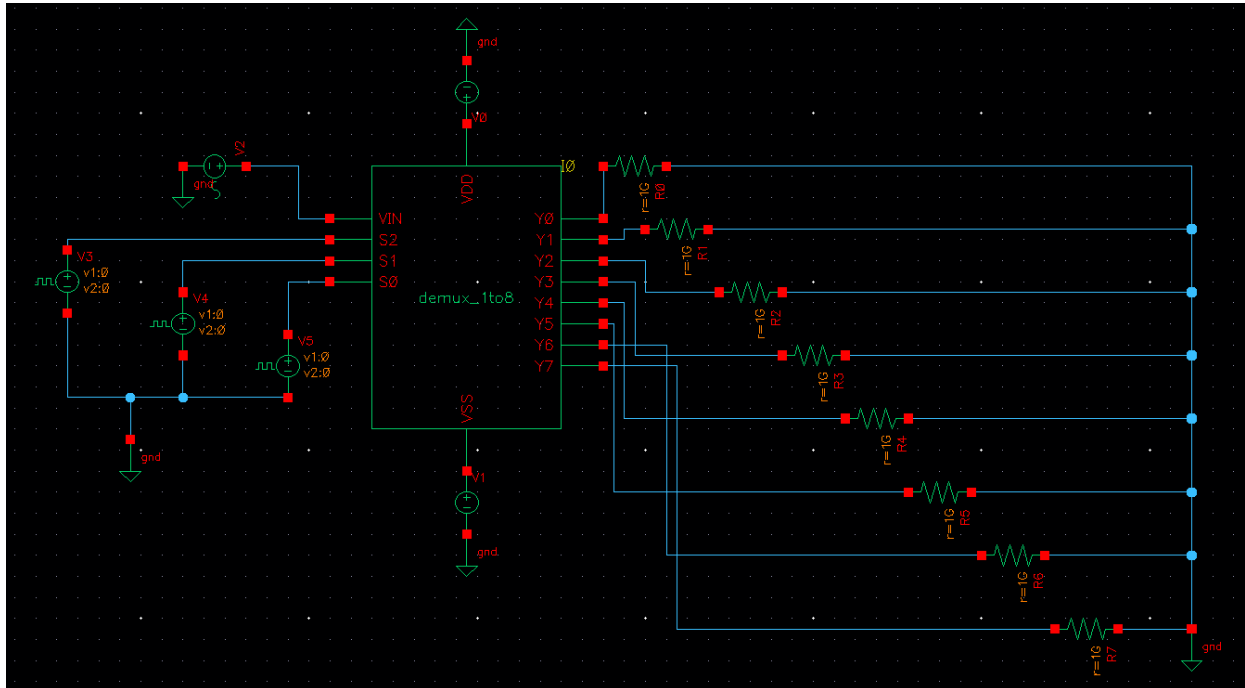


Figure 8. Functional testbench used to cycle through all eight output selections.

Time interval	S2S1S0	Selected output
0-10 μ s	000	Y0
10-20 μ s	001	Y1
20-30 μ s	010	Y2
30-40 μ s	011	Y3
40-50 μ s	100	Y4
50-60 μ s	101	Y5
60-70 μ s	110	Y6
70-80 μ s	111	Y7

4. Functional Validation

4.1 Eight-output routing

The transient response confirms that the input is delivered sequentially to Y0-Y7 according to the binary select code. When all outputs are plotted together, the selected trace often lies directly on top of VIN, while small residual pieces can remain visible on unselected outputs because of MOS parasitic capacitances and clock feedthrough.

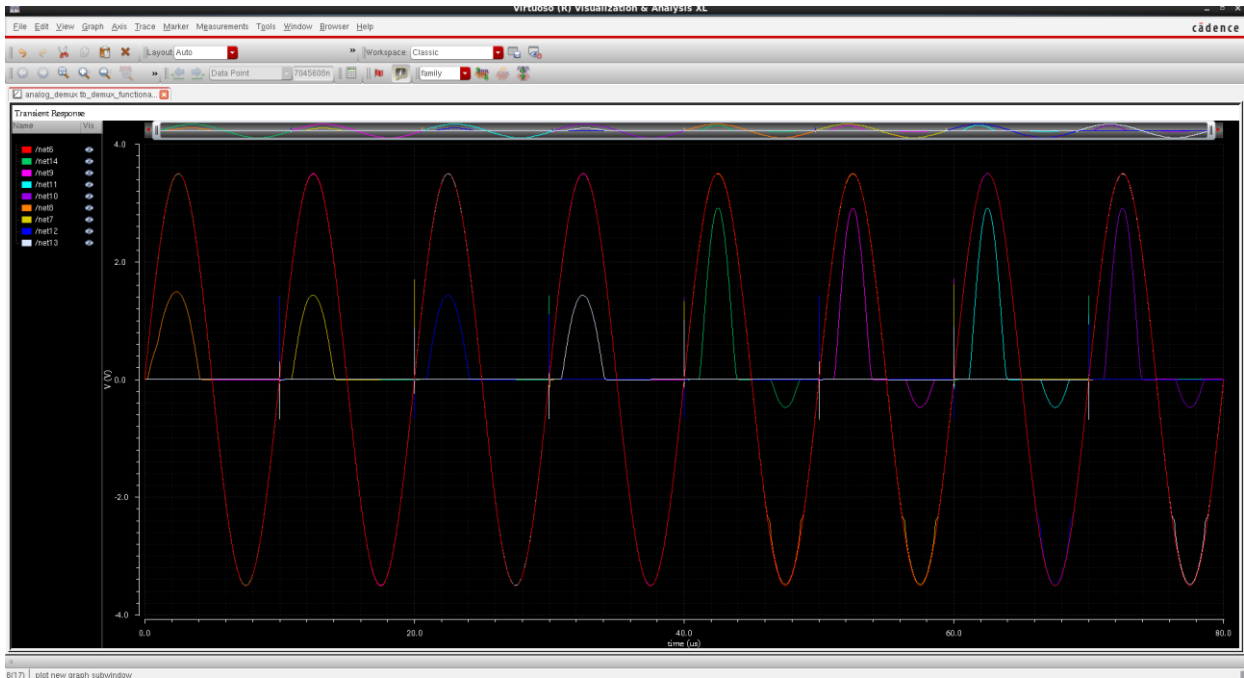


Figure 9. Final 80 μ s functional regression: all eight selections operate in the expected sequence.

4.2 Fixed-code proof that selected paths have equal gain

To prove that the apparent size difference between the first four and last four colored traces was not an ON-state gain error, fixed-code transient tests were performed. With code 000, Y0 overlaps VIN; with code 100, Y4 also overlaps VIN. This confirms that the selected upper and lower halves of the tree have essentially the same signal transfer.

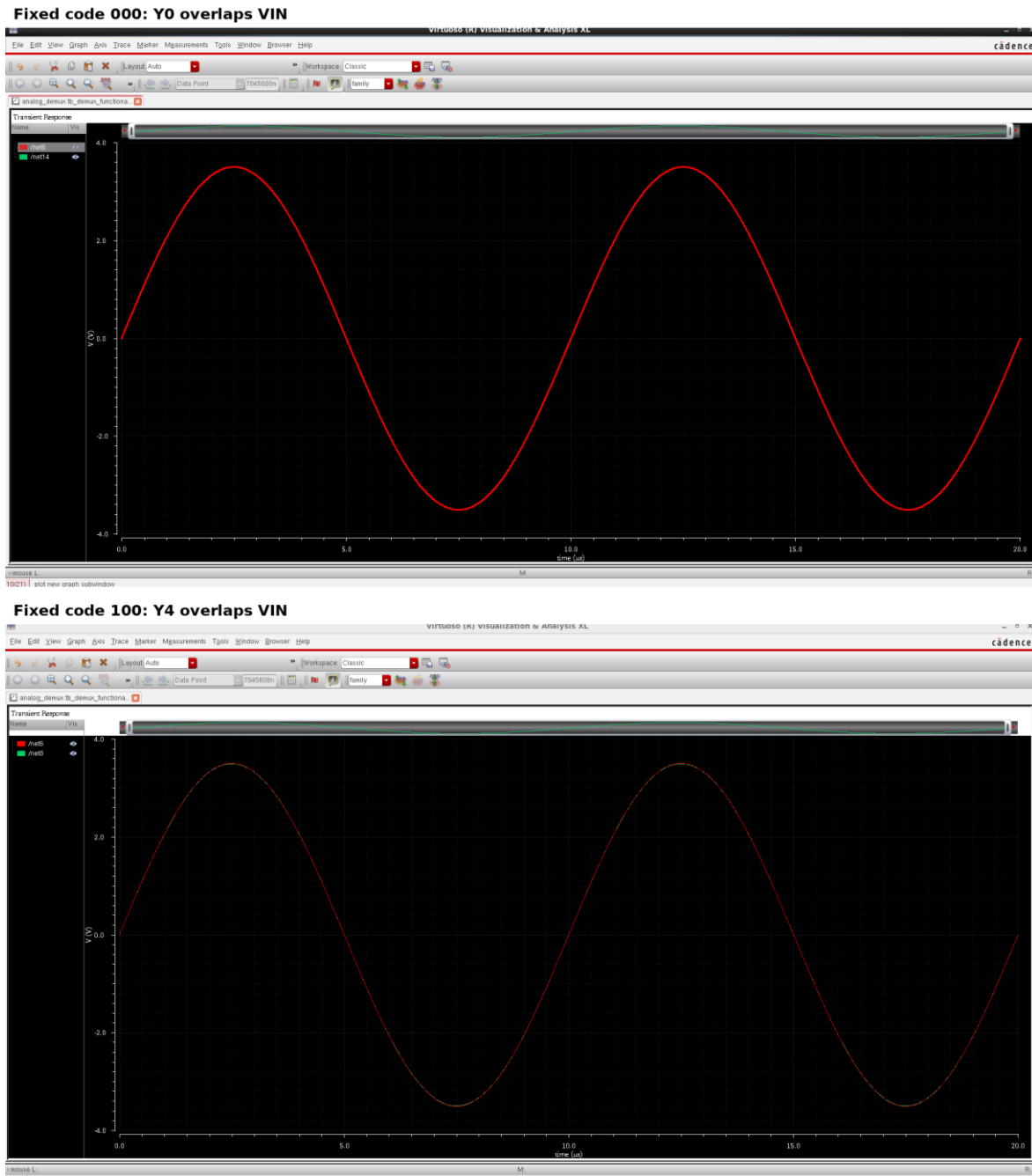


Figure 10. Fixed-code tests showing Y0 and Y4 overlapping the input waveform when selected.

4.3 OFF-output feedthrough

With extremely weak $1\text{ G}\Omega$ output pull-downs, an unselected output could show a small waveform (for example, a few hundred millivolts in an early Y1 observation). Reducing the functional-test pull-down to $1\text{ M}\Omega$ and then $10\text{ k}\Omega$ reduced this residual signal strongly, proving that it was parasitic capacitive feedthrough rather than an unintended ON path. The $10\text{ k}\Omega$ functional load was retained only for clean visualization; specification-specific tests use their required loads.

5. Verification of the Electrical Criteria

Each specification was checked with a dedicated simulation condition rather than trying to extract all parameters from one testbench. The load and excitation for each test are stated explicitly below so the measurements are reproducible.

5.1 Analog signal range and complete-path ON resistance

Code 000 was fixed so Y0 was selected. VIN was set to +3.5 V DC and Y0 was loaded by 1 kΩ. The measured Y0 value was approximately 3.41627 V. The complete three-stage on-resistance follows from the voltage divider formed by RON and the 1 kΩ load:

$$R_{ON} = R_L (V_{IN} / V_{OUT} - 1)$$

$$R_{ON}(+3.5 \text{ V}) = 1000(3.5 / 3.41627 - 1) \approx 24.51 \Omega$$

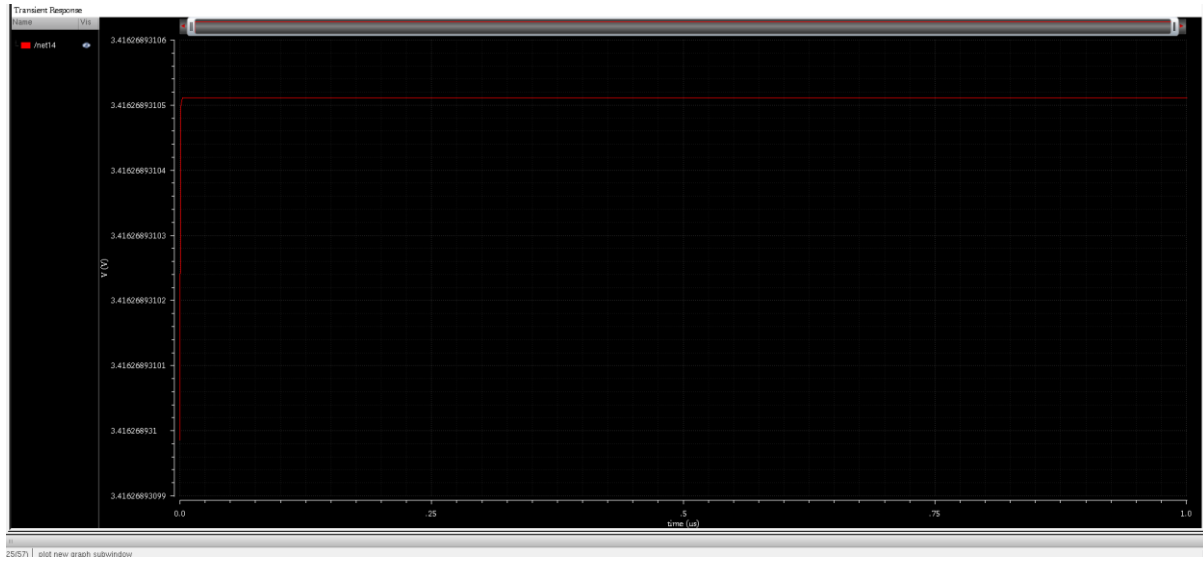


Figure 11. +3.5 V selected-path output used for the full-tree ON-resistance calculation.

Repeating at VIN = -3.5 V gave VOUT ≈ -3.42517 V and RON ≈ 21.85 Ω. The worst of the two endpoints is therefore 24.51 Ω, just under the 25 Ω limit. A DC transfer sweep also demonstrated signal transfer across the negative and positive analog range.

$$R_{ON}(-3.5 \text{ V}) = 1000(3.5 / 3.42517 - 1) \approx 21.85 \Omega$$

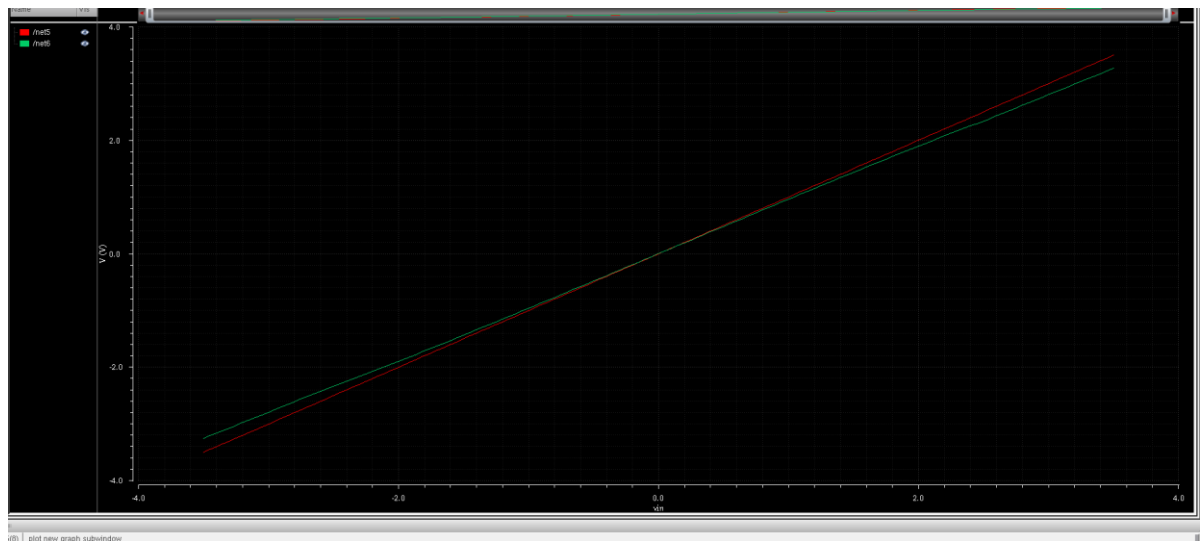


Figure 12. DC transfer through the selected path over the analog input range.

Result: Worst-case measured full-path RON ≈ 24.51 Ω. The 10-25 Ω criterion is satisfied, although the positive endpoint is close to the upper limit. The ±3.5 V analog range also passes.

5.2 Input capacitance

Input capacitance was measured by driving VIN with an AC source of magnitude 1 V, placing a current probe in series, and reading the input-current magnitude near 1 MHz. For a capacitive input,

$$C_{IN} = |I_{IN}| / (2\pi f |V_{AC}|)$$

The final measurement marker was $f = 1.006643 \text{ MHz}$ and $|I_{IN}| = 348.3155 \text{ } \mu\text{A}$. With $V_{AC} = 1 \text{ V}$:

$$C_{IN} = 348.3155 \text{ } \mu\text{A} / [2\pi(1.006643 \text{ MHz})(1 \text{ V})] \approx 55.07 \text{ pF}$$

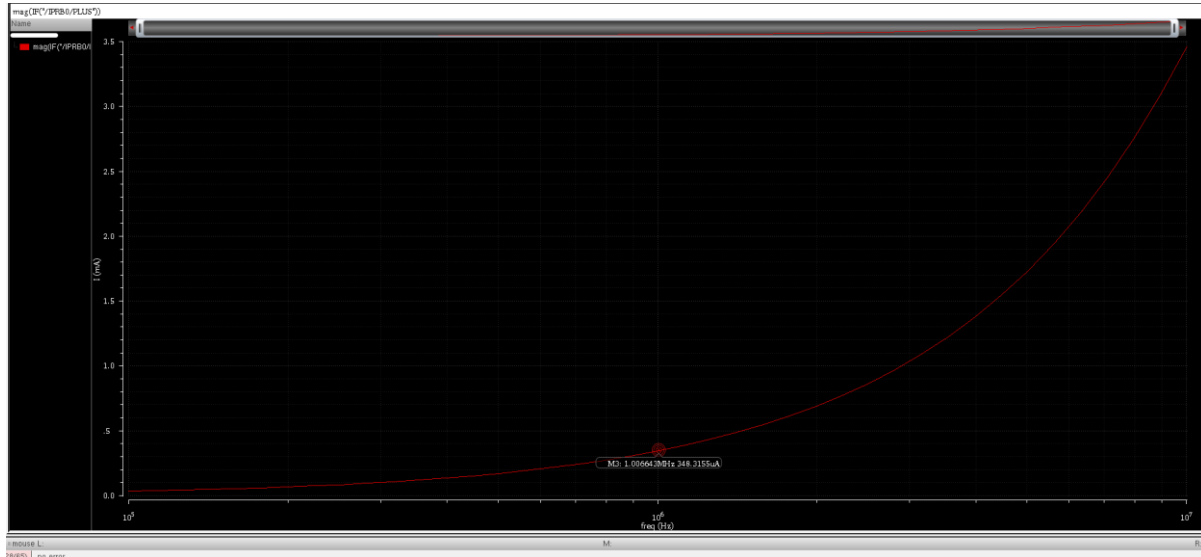


Figure 13. AC input-current measurement used to obtain $C_{IN} \approx 55.07 \text{ pF}$.

Result: Final $C_{IN} \approx 55.07 \text{ pF}$, which lies inside the project's stated 50-65 pF band.

5.3 Switching turn-ON time

For the switching-time test, Y1 was selected by changing S0 from 0 V to 2 V while S2 = S1 = 0 V. VIN was held at +3.5 V. The required load was applied directly: RL = 1 kΩ in parallel with CL = 10 pF. The timing reference was the 50% point of S0 (1 V), and the output point was 90% of its final value.

$$t_{\text{ON}} = t[Y1 = 0.9V_{\text{final}}] - t[S0 = 1 \text{ V}]$$

$$t_{\text{ON}} = 101.09097 \text{ ns} - 100.50016 \text{ ns} \approx 0.59081 \text{ ns}$$

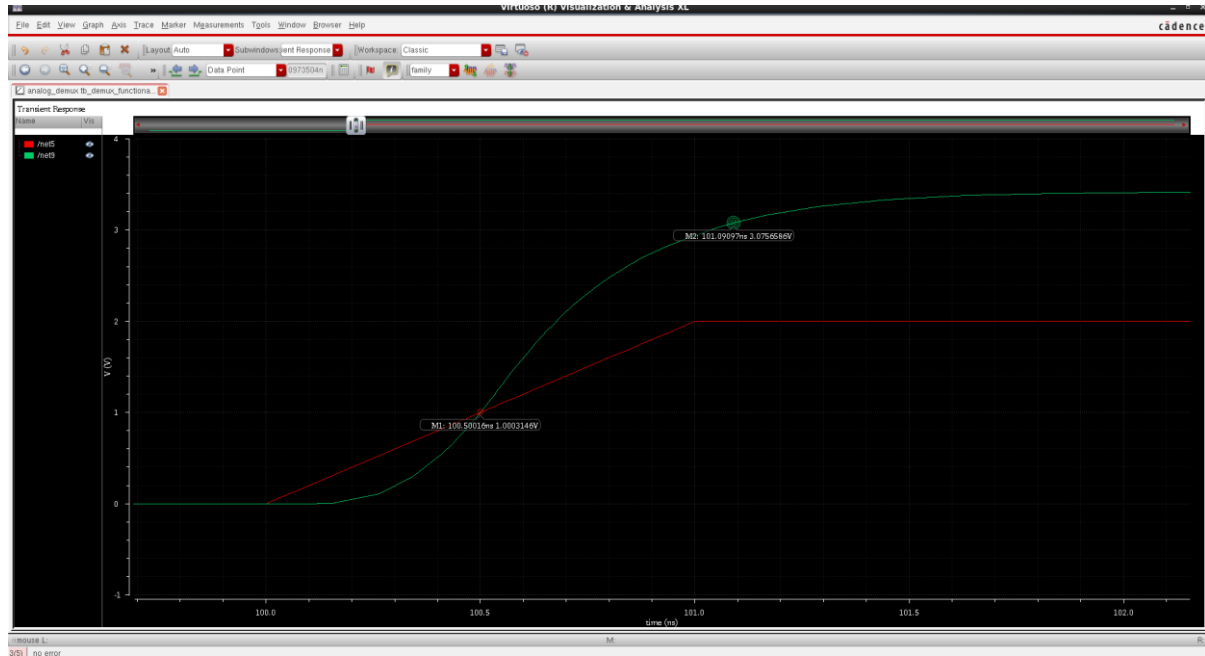


Figure 14. Marker-based turn-ON time measurement with RL = 1 kΩ and CL = 10 pF.

Result: tON ≈ 0.591 ns, far below the 120 ns maximum.

5.4 -3 dB bandwidth

With code 000 fixed, an AC sweep was performed with Y0 loaded by 1 kΩ || 10 pF. The plotted transfer expression was 20log10|VY0/VIN|. The low-frequency insertion loss was approximately -0.123 dB. The true 3 dB-down level relative to that baseline is therefore about -3.123 dB.

$$\text{Gain}_{\text{dB}} = 20 \log_{10} |V_{\text{OUT}} / V_{\text{IN}}|$$

$$3 \text{ dB-down level} \approx -0.123 \text{ dB} - 3 \text{ dB} = -3.123 \text{ dB}$$

The response was extended to 1 GHz. Even at 1 GHz the curve was only around -3.0 dB, so the exact -3.123 dB crossing is above 1 GHz. The instructor clarified that an actual 1.2 MHz cutoff was preferred, but if that could not be achieved, a -3 dB bandwidth greater than 1.2 MHz was acceptable. The final design therefore satisfies the allowed condition.

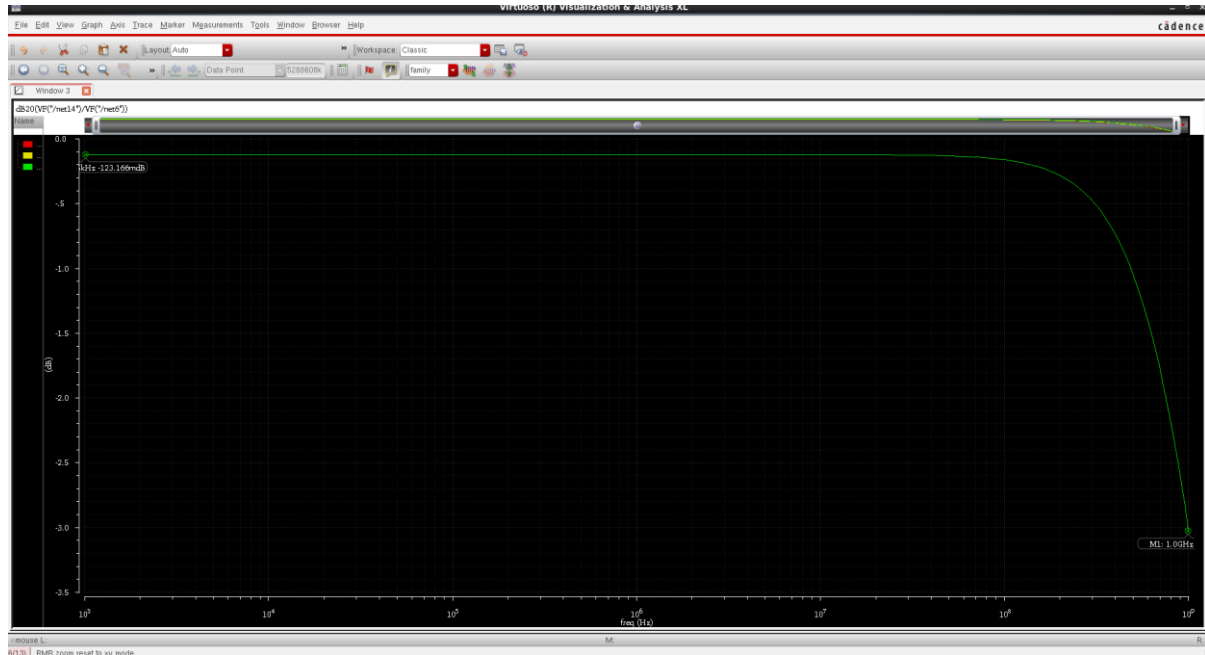


Figure 15. Frequency response showing that the -3 dB bandwidth is above 1 GHz.

Result: $f_{-3dB} > 1 \text{ GHz}$. This is much greater than 1.2 MHz and meets the instructor-approved fallback condition, but the actual cutoff is not centered at 1.2 MHz.

5.5 Charge injection over the full signal swing

Charge injection was measured during an ON-to-OFF switching event because stored channel charge is released when the transmission gate turns OFF. Y1 was initially selected with $S_0 = 2 \text{ V}$ and then switched OFF with S_0 falling to 0 V. The Y1 load was $C_L = 10 \text{ pF}$ in parallel with $1 \text{ G}\Omega$ so the injected charge appeared as a small held voltage step.

$$Q_{inj} = C_L |V_{after} - V_{before}|$$

At $V_{IN} = 0 \text{ V}$, the final step was approximately 28.65 mV. At +3.5 V the step was approximately 23.507 mV. At -3.5 V the step was approximately 4.875 mV. These give:

V_{IN}	$ \Delta V $	$Q_{inj} = 10 \text{ pF} \times \Delta V $
0 V	28.65 mV	0.2865 pC
+3.5 V	23.507 mV	0.2351 pC
-3.5 V	4.875 mV	0.0487 pC

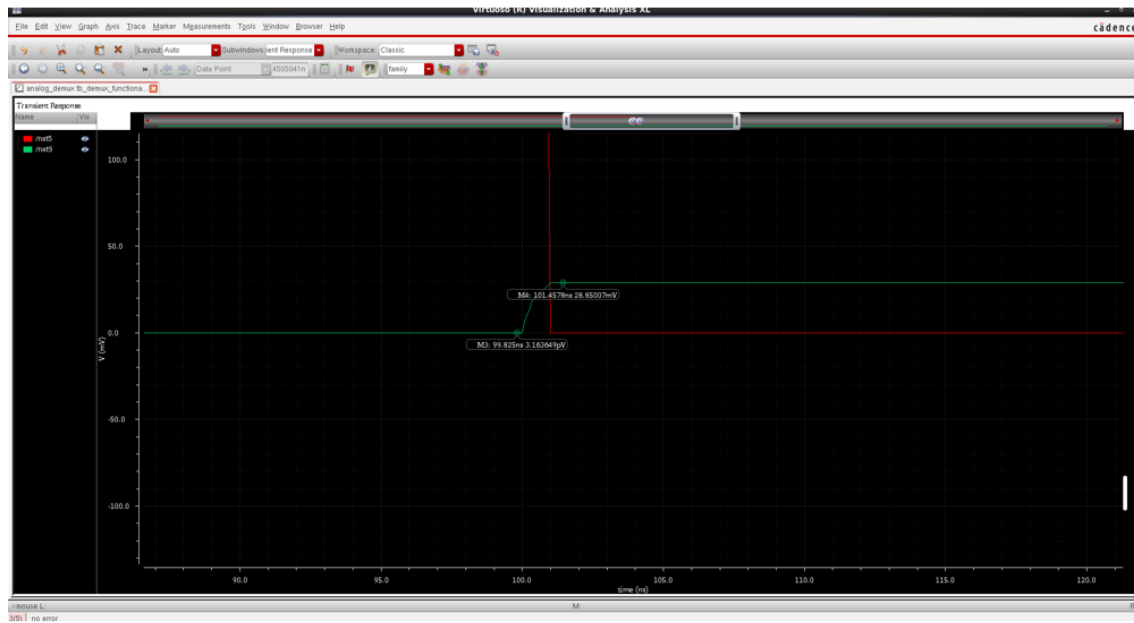
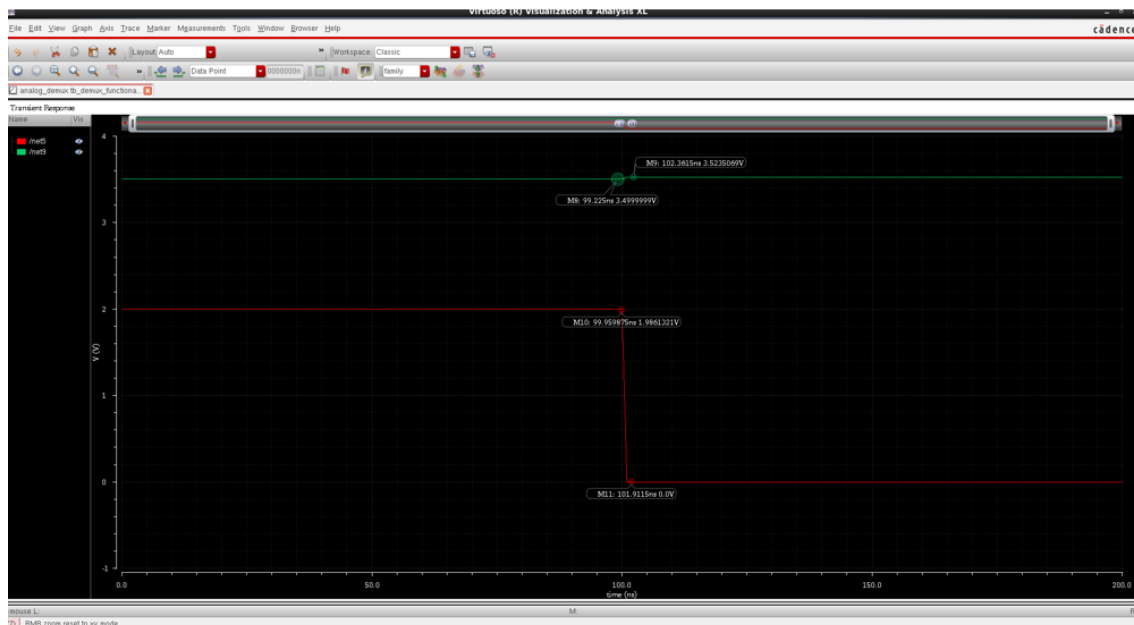
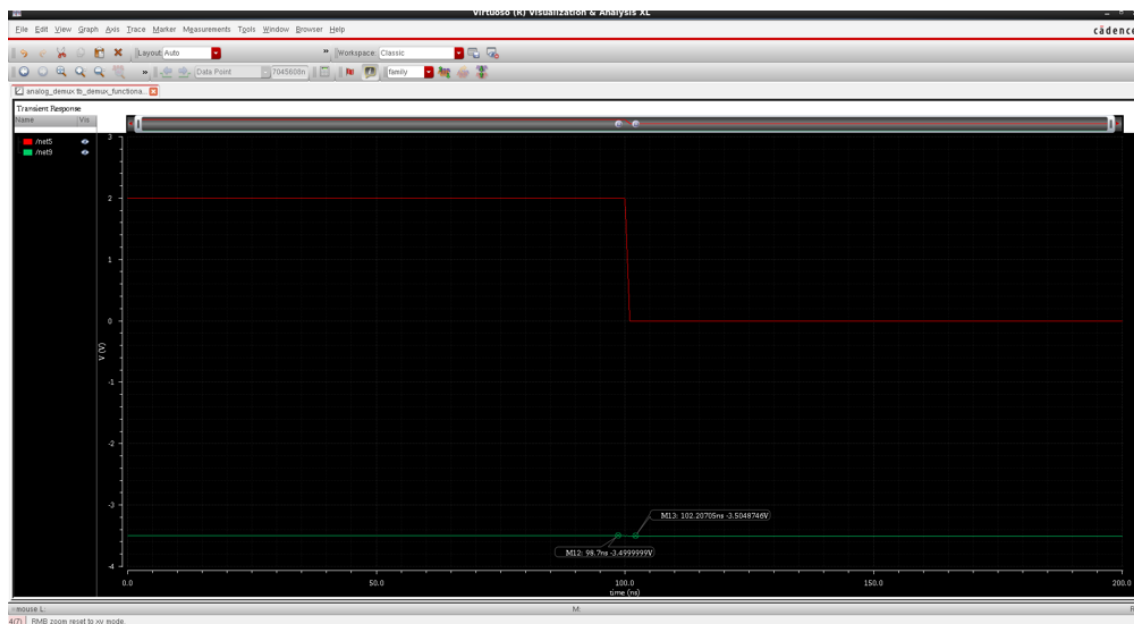
VIN = 0 V**VIN = +3.5 V****VIN = -3.5 V**

Figure 16. Charge-injection measurements at $V_{IN} = 0\text{ V}$, $+3.5\text{ V}$ and -3.5 V .

Result: Worst measured charge injection $\approx 0.287\text{ pC}$, well below the 20 pC maximum.

5.6 Power dissipation

Average supply currents were measured during the original $80\text{ }\mu\text{s}$ functional switching sequence, with a 100 kHz , $\pm 3.5\text{ V}$ sine input and all select bits cycling. A 0.5 ns maximum transient step was used to capture narrow switching-current spikes. Cadence returned the following average source currents:

$$I_{\text{VDD,avg}} = +243.7\text{ nA}$$

$$I_{\text{VSS,avg}} = -205.2\text{ nA}$$

Using the magnitudes of the two $\pm 4\text{ V}$ rail contributions:

$$P_{\text{total}} \approx 4(243.7\text{ nA}) + 4(205.2\text{ nA}) \approx 1.7956\text{ }\mu\text{W} \approx 0.00180\text{ mW}$$

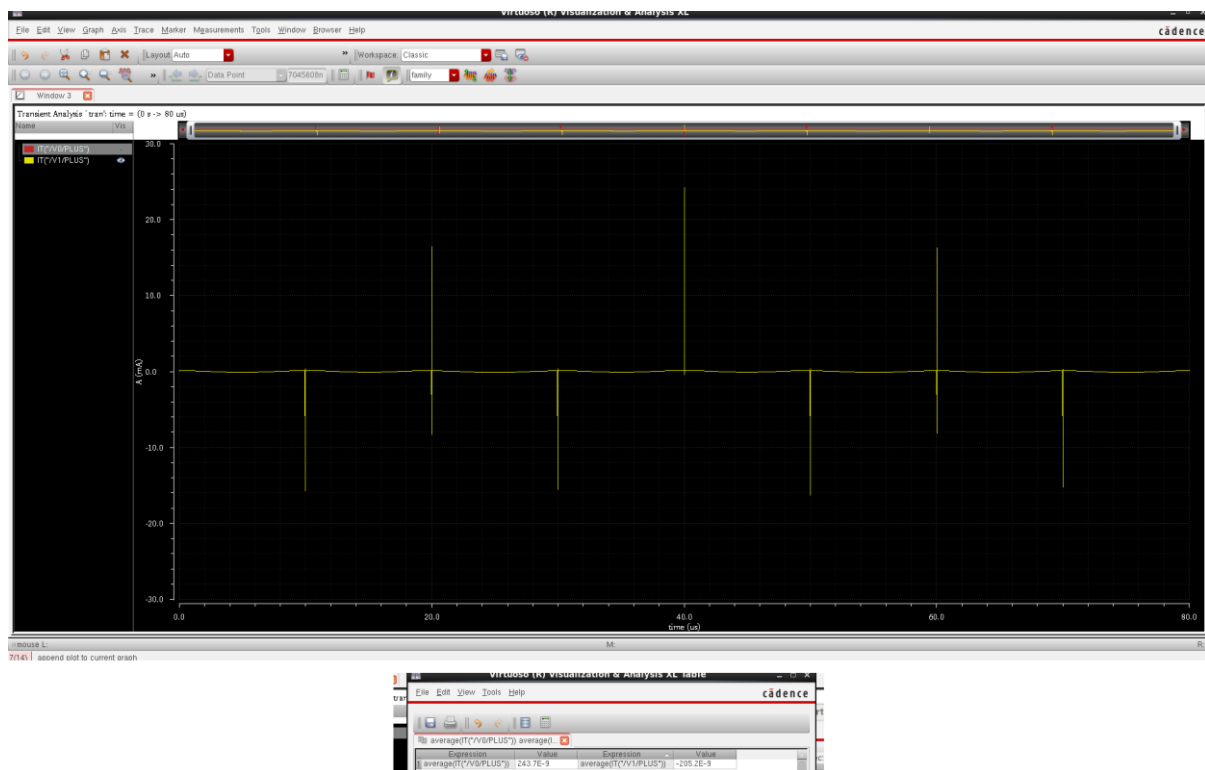


Figure 17. Supply-current switching spikes and the average-current values used for the power estimate.

Result: Simulated DEMUX/core rail power $\approx 1.80\text{ }\mu\text{W}$, far below the 25 mW maximum. Because the internal level converters are ideal VCVSs, this is not a complete physical-chip power number.

6. Problems Encountered and How They Were Solved

Problem / observation	Resolution and engineering interpretation
High resistance through three cascaded switches	A selected 1:8 path contains three transmission gates in series. Early sizing could have pushed the total above 25 Ω . The NMOS/PMOS widths were increased using multiple fingers, and the complete path was measured under a 1 k Ω load. Final worst-case RON was $\approx 24.51 \Omega$.
External 0/2 V logic could not directly provide the ± 4 V complementary TG control	Two ideal VCVS equations per select bit were used to produce $SH = 4S - 4$ and $SHB = 4 - 4S$. This preserved the required external logic levels while allowing strong transmission-gate control.
Ground-net naming conflict in Cadence	The analogLib ground symbol creates global net gnd!. Applying another “ground” label to the same wire caused a short/name conflict. The solution was to use analogLib/gnd consistently and avoid a competing local ground label.
OFF outputs appeared to contain small analog waveforms	With 1 G Ω output pull-downs, parasitic capacitive feedthrough was visible on unselected outputs. Changing the functional pull-down to 10 k Ω made the OFF outputs settle close to 0 V. Fixed-code Y0 and Y4 tests proved that selected-path gain remained correct.
DC on-resistance test initially produced an incorrect ramp-looking waveform	The wrong stimulus/analysis combination made the plotted node appear to ramp. The test was simplified to a true DC input (vdc) and steady-state output measurement, after which the correct ~ 3.416 V selected output was obtained.
Input capacitance was much lower than the stated 50-65 pF target	The intrinsic result was ~ 2.05 pF. Approximately 53 pF was added inside demux_1to8 from VIN to VSS, and the AC-current test was repeated. Final CIN became ~ 55.07 pF.
Bandwidth plot initially looked like -123 dB	The y-axis was actually in mdB (milli-decibels): $-123 \text{ mdB} = -0.123 \text{ dB}$. After recognizing the unit and extending the sweep to 1 GHz, the response was correctly interpreted as a very wide-band path.
Ambiguity in the 1.2 MHz bandwidth requirement	The instructor clarified that an actual 1.2 MHz cutoff was preferred, but explicitly allowed -3 dB bandwidth $> 1.2 \text{ MHz}$ if exact tuning could not be achieved. The final $> 1 \text{ GHz}$ result therefore meets the accepted condition.
Charge injection needed to be separated from ordinary turn-on behavior	The measurement used an ON-to-OFF 2 V $\rightarrow$ 0 V control transition and a 10 pF hold capacitor, so the residual output step directly represented injected charge. Three VIN values were checked to cover the full signal swing.
Power result could be misleadingly small	The transmission-gate core has very low static current, and ideal VCVS level conversion does not consume realistic control power. The report therefore labels the 1.80 μW result as simulated core/rail power rather than complete physical implementation power.

7. Final Performance Summary

Criterion	Requirement	Measured / achieved	Status
Supply voltage	$\pm 4 \text{ V}$	$\pm 4 \text{ V}$	PASS
Logic high / low	2 V / 0 V	2 V / 0 V	PASS
Analog range	-3.5 to +3.5 V	Verified across $\pm 3.5 \text{ V}$	PASS
ON resistance	10-25 Ω	24.51 Ω worst case	PASS
Input capacitance	50-65 pF target	55.07 pF	PASS
Turn-ON time	$\leq 120 \text{ ns @ } 1 \text{ k}\Omega \parallel 10 \text{ pF}$	0.591 ns	PASS
Charge injection	$\leq 20 \text{ pC}$ (spec line: 1-20 pC max)	0.287 pC worst case	PASS
Power dissipation	$\leq 25 \text{ mW}$ (spec line: 1-25 mW max)	$\approx 1.80 \mu\text{W}$ core/rail estimate	PASS*
Bandwidth, -3 dB	Preferred $\approx 1.2 \text{ MHz}$; fallback $> 1.2 \text{ MHz}$	$> 1 \text{ GHz}$	PASS (fallback)
Functional routing	Correct Y0-Y7 selection	All eight codes verified	PASS

*Power is the simulated DEMUX/core rail power with ideal VCVS level conversion. It is not a full transistor-level level-shifter power measurement.

8. Limitations and Design Notes

- The control-level conversion is behavioral/ideal. A physical 0/2 V to ± 4 V level shifter would add transistors, delay, area and power and must be designed with suitable high-voltage devices or stacking techniques.
- The selected gpdk090 nmos2v/pmos2v devices are low-voltage model devices; use of the ± 4 V control environment is an academic schematic-level assumption and is not a reliability-qualified physical implementation.
- No layout, parasitic extraction, DRC or LVS was performed. Post-layout parasitic capacitances would reduce the very high simulated bandwidth and may change timing and charge injection.
- The functional 10 k Ω pull-downs are visualization loads. The official parameter tests used the loads specified for each measurement.
- The measured bandwidth satisfies the instructor-approved >1.2 MHz fallback but does not implement an actual 1.2 MHz cutoff.
- The final CIN was deliberately tuned into the stated 50-65 pF band. If “max 50-65 pF” is interpreted only as an upper limit, the untuned intrinsic value was already compliant.

9. Conclusion

A complete 1:8 analog demultiplexer was successfully designed and simulated using a modular CMOS transmission-gate architecture. The seven-cell binary tree provides a clear and scalable implementation: one 1:2 stage controlled by S2, two by S1, and four by S0. Each 1:2 cell is built from two complementary CMOS transmission gates, giving a bidirectional analog path capable of passing the required ± 3.5 V input range.

The final functional simulations confirmed correct routing for all eight select codes. Separate measurement setups verified the major electrical criteria: worst-case full-path on-resistance $\approx 24.51 \Omega$, input capacitance ≈ 55.07 pF, turn-ON time ≈ 0.591 ns, worst charge injection ≈ 0.287 pC, and simulated core power $\approx 1.80 \mu\text{W}$. The -3 dB bandwidth is above 1 GHz and therefore meets the instructor-approved condition of remaining above 1.2 MHz when an exact 1.2 MHz cutoff is not obtained.

The main remaining step for a production-grade implementation would be replacing the ideal level conversion with a physical high-voltage-compatible control circuit and then repeating verification after layout and parasitic extraction. Within the stated schematic-and-simulation scope, the project requirements have been demonstrated with numerical and waveform evidence.

Appendix A - Verification Test Matrix

Test	Input	Control	Load	Analysis	Measured quantity
Functional routing	VIN: 100 kHz sine, 3.5 V amplitude	S0/S1/S2 pulses covering 000-111	10 kΩ on Y0-Y7	80 μs; maxstep ~20 ns	Correct output sequence
ON resistance	VIN = +3.5 V and -3.5 V DC	000	Y0: 1 kΩ; others weak	DC/steady transient	VOUT and divider equation
Input capacitance	AC magnitude = 1 V	000	Outputs weakly loaded	AC; marker ~1 MHz	IIN → CIN
Turn-ON time	VIN = +3.5 V DC	S0: 0→2 V; S2=S1=0	Y1: 1 kΩ 10 pF	Transient; maxstep 0.5 ns	50% select to 90% output
Bandwidth	AC magnitude = 1 V	000	Y0: 1 kΩ 10 pF	AC 1 kHz to 1 GHz	$20\log_{10} V_{Y0}/V_{IN} $
Charge injection	VIN = 0, +3.5, -3.5 V	S0: 2→0 V	Y1: 10 pF 1 GΩ	Transient; maxstep 0.05 ns	$Q = CL\Delta V$
Power	100 kHz sine, ±3.5 V	All bits switching	Functional loads	80 μs; maxstep 0.5 ns	Average VDD and VSS source currents

Appendix B - Key Equations

$$R_{ON} = R_L (V_{IN} / V_{OUT} - 1)$$

$$C_{IN} = |I_{IN}| / (2\pi f |V_{AC}|)$$

$$t_{ON} = t(\text{output at 90\% final}) - t(\text{select at 50\%})$$

$$\text{Gain}_{dB} = 20 \log_{10} |V_{OUT} / V_{IN}|$$

$$Q_{inj} = C_L |V_{after} - V_{before}|$$

$$P_{total} \approx |V_{DD} I_{DD,avg}| + |V_{SS} I_{SS,avg}|$$

$$S_H = 4S - 4, \quad S_{HB} = 4 - 4S$$